

Yuval Munz

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SUMMARY

I enter unfamiliar systems, work out how they run, and find where small design decisions create friction, wasted effort, or missed opportunity. I break those systems down, question the assumptions behind them, and build the more efficient version.

EDUCATION

Queen's University

Bachelor of Applied Science, Mechanical Engineering

Kingston, Ontario

Class of 2028

TECHNICAL SKILLS

CAD & Simulation: Solidworks (part modelling, Simulation FEA), static structural analysis, load case definition, topology and mass optimization

Analysis & Modelling: MATLAB (lap simulation, drivetrain analysis), Python (steady-state thermal modelling), Excel modelling, first-principles hand calculation

Design & Manufacturing: Material selection (7075-T6, steels, insulation), DFM for waterjet, 3D printing and rapid prototyping, ergonomic design from user requirements

Powertrain: Chain and sprocket drive design, gear ratio optimization, EV powertrain systems (EMRAX 188 HV)

Software & Automation: Python, Workato, agentic workflow design, LLM and API integration (Slack, Jira)

EXPERIENCE

Powertrain Engineer, Gear Reduction & Sprockets Lead

Queen's Formula SAE Racing Team

Sept 2024 – Present

Kingston, Ontario

- Own chain and sprocket drive design for a 58:12 (4.83:1) #530 final drive on an EMRAX 188 HV powertrain (100 Nm, 1000–8000 RPM), sizing chain tension from peak motor torque and validating in Solidworks FEA on 7075-T6.
- Directing a first-year engineer through a full design cycle: sizing, FEA, manufacture, and validation.

Product Intern

Workato

March 2026 – Present

San Francisco, USA (Remote)

- Built a Genie agent in Slack to triage support load during beta of AIRO, routing channel messages across four skills: logging questions, capturing bugs, filing Jira tickets, and answering from a knowledge base that self-updated by escalating unanswered questions to the team and folding their replies back in. Deflected ~30 daily requesters off a 4-person product team; still running post-GA.
- Stood up an in-house branded animation and graphics pipeline for product demo videos when design capacity was unavailable, producing 20 videos published to Workato's YouTube channel.

Mechanical Project Assistant Intern

Green Air LTD

May 2025 – Aug 2025

Paphos, Cyprus

- Diagnosed and repaired mechanical faults in an independent, hands-on service environment.

IT Support Intern

CMiC

May 2024 – Sept 2024

Toronto, Ontario

- Coordinated a company-wide Mac to Windows laptop migration across 50 employees, driving outreach and tracking transition status to completion; provisioned software, onboarded new hardware, and performed component-level repairs.

3D Printing & Fabrication Intern

3D Factory

Nov 2023 – May 2024

- Worked with clients and designers to translate CAD models into physical components, ensuring design for manufacturability.

PROJECTS

- Sprocket Topology & Mass Optimization** | *Solidworks, FEA, MATLAB, Waterjet* Nov 2025 – Present
- Cut rear sprocket mass 30% (520 g to 364 g) via waterjet cutouts in 7075-T6, holding peak deflection to 0.159 mm under 2,700 N chain load.
 - Inverted the method after Solidworks topology tools could not hold rotational symmetry across a 6-bolt pattern: benchmarked 3 cutout patterns over 12 FEA iterations at fixed mass budget to find the load-path geometry maximizing stiffness per gram.
 - Building a MATLAB lap simulation (OpenLAP) to evaluate final drive ratios against lap time, sweeping ratios across the EMRAX 188 HV torque curve to quantify the acceleration vs top-speed trade-off.
 - Diagnosed a chain wear failure mode where a 110-link chain on a 55T sprocket divided evenly, cycling every tooth against the same two rollers all season; resolved by moving to 56T to break the common factor.
- Cabin Zoning Concept for Rivian R1S Camp Mode** | *Python, Thermal Modelling, CAD* Oct 2025 – Jan 2026
- Identified an EV energy-optimization opportunity: Camp Mode conditions the full cabin while occupants use only the rear ~60%, spending HVAC load on unoccupied volume.
 - Built a steady-state heat-loss model in Python ($Q = U \cdot A \cdot \Delta T$) across cabin surfaces, estimating a 40% reduction in HVAC draw from conditioning the rear zone alone.
 - Specified a lightweight insulated partition on thermal conductivity and mass, and produced concept renders of the zoned cabin.
- Off-Road Racing Steering Wheel** | *Baja SAE, CAD, FEA, Ergonomic Design* Jan 2025 – Apr 2025
- Designed a steel off-road steering wheel from driver requirements, molding grip geometry to interviewed drivers' hand profiles for grip retention over rough terrain.
 - Validated in FEA under two load cases: axial loading at the bolt pattern and asymmetric rotational input representing opposed-hand steering force.
 - Produced a printed prototype for fit and ergonomic review ahead of steel manufacture.

VENTURES

- Technical Co-Founder & CTO** April 2026 – Present
- Pursr* | *SR&ED Evidence Infrastructure for Canadian Manufacturing SMEs*
- Identified that SR&ED documentation is reconstructed from memory months after the engineering work happens, and built a platform that captures contemporaneous, audit-ready evidence at the point of work instead.
 - Scoped CNC machine-data integration on MTConnect rather than FOCAS, choosing the read-only protocol because bidirectional write access is unacceptable to plant IT and machine tool builders.
 - Deployed a working platform and repositioned it consultant-first (firm → client → project tenancy) after validating demand with a Toronto SR&ED consulting firm; built a Slack round-trip so a documentation gap becomes a prompted question and the engineer's reply lands as structured evidence.
- Founding Engineer & CTO** April 2026 – Present
- Actiograph* | *Application-Tracking Platform for Canadian Pre-Med Students*
- Identified a timing failure in medical school applications: students begin assembling their record months before applying, when the experiences it draws on span the prior five years. Built a system that captures each experience, reflection, and verifier contact as it happens.
 - Built and shipped the React Native application to TestFlight, replacing the scattered notes, spreadsheets, and documents students currently use with a single structured record.
 - Ran structured testing with an advisory board of pre-med and medical students; prepared and submitted the DDQIC Ignite qualifying pitch at Queen's University.